Eko AI-Assisted Partner Performance Data Worker

Final Project Documentation — Data Analytics / BI / AI Workflow Internship Evaluation

1. Goal

- Build an AI-assisted Data Worker that analyzes partner business data, identifies performance signals, classifies partners, prioritizes attention, generates recommended actions, validates input quality, and escalates exceptions for human review.

2. User

- Business, operations, and analytics teams who need to quickly identify partners requiring attention and understand the recommended next action.

3. System

- Business Data → Analysis → Classification → Priority → Action Recommendation → Validation → Proceed to Output / Human Review

4. Inputs

- Partner ID
- Current 30D GMV
- Previous 30D GMV
- Current 30D Transactions
- Previous 30D Transactions
- Total Transactions
- Total GMV
- Last Transaction Date
- Days Since Last Transaction

5. Outputs

- GMV Growth %
- Partner Classification
- Priority
- Recommended Action
- Validation Status
- Validation Issue
- Validation Message
- Human-review escalation

6. Decisions

- Classify partners as Improving, Declining, Active, Inactive, Risky, or High Potential.
- Assign High, Medium, or Low priority.
- Recommend an action based on business signals.
- Determine whether input data passes validation.
- Escalate records requiring human review.

7. Constraints

- The current implementation is a rule-based decision engine built in Power BI/DAX. It does not alter source data or replace human judgment when validation fails.

8. Definition of Done

- Business data is analyzed; partners are classified and prioritized; an action-oriented output is produced; validation checks are completed; exceptions are identified; and cases requiring human judgment are escalated.

9. Escalation

- The worker requests human review when required inputs are missing or a validation rule identifies an exception.

10. Success Metrics

- Primary: reduce manual effort needed to identify partners requiring attention and determine next actions.
- Operational: percentage of records processed automatically without human intervention.
- Quality: percentage of exception records correctly detected and escalated.

11. Data Dictionary

Field	Meaning	Type
Partner_ID	Unique partner identifier	Text
Current_30D_GMV	GMV generated in current 30-day period	Numeric
Previous_30D_GMV	GMV generated in previous 30-day period	Numeric
GMV_Growth_%	Percentage change in GMV	Numeric
Current_30D_Transactions	Current-period transaction count	Integer
Previous_30D_Transactions	Previous-period transaction count	Integer
Total_Transactions	Total transactions for partner	Integer
Total_GMV	Total GMV for partner	Numeric
Last_Transaction_Date	Date of most recent transaction	Date
Days_Since_Last_Transaction	Days since most recent transaction	Integer
Classification	Partner performance segment	Text
Priority	Level of attention required	Text
Recommended_Action	Suggested business action	Text
Validation_Issue	Detected data-quality issue	Text
Validation_Status	Valid or Requires Human Review	Text
Validation_Message	Explanation of validation result	Text

12. Assumptions

- The dataset contains dummy/sample business data.
- GMV Growth is based on current versus previous 30-day GMV.
- Classification and priority are rule-based.
- Median GMV Growth is used in the executive dashboard because extreme values can distort the average.
- GMV Growth above 1000% is treated as an invalid/extreme value requiring review.
- Missing GMV Growth prevents the worker from proceeding with a recommendation.
- Validation thresholds are prototype assumptions and require business-owner approval before production use.

13. Current Version vs Next Version

- Current Version — Rule-based decision engine: Power BI/DAX classification, priority, validation, and structured action recommendations.
- Next Version — LLM-powered workflow: an LLM could interpret validated signals, generate contextual action notes, explain anomalies, and automate workflow handoffs while retaining deterministic validation and human escalation controls.

14. Intentional Failure Scenario

- EKO10250 was intentionally introduced with a missing validated GMV Growth input. The worker flags it as Requires Human Review and prevents the recommendation workflow from proceeding automatically.

- Validation message: “GMV Growth is missing. Human review required before recommendation.”

15. Validation & Exception Handling — Current Scope

- Implemented and demonstrated: missing GMV Growth detection using the intentional EKO10250 failure case.
- Implemented and demonstrated: extreme GMV Growth detection using the >1000% prototype threshold.
- Implemented: duplicate Partner ID detection logic in the validation layer.
- Not intentionally injected into the current 250-row validation dataset: a duplicate-record failure scenario. The duplicate rule is documented as available, but no duplicate test record was left in the final validation dataset.
- The Validation & Exception Center currently validates the full 250-row dataset and displays Valid vs Requires Human Review, issue, and message.

16. Power BI Prototype

- Executive Overview: portfolio KPIs, classification distribution, priority distribution, GMV analysis, and transaction recency.
- Partner Detail & AI Action Center: partner-level KPIs, performance details, and structured action recommendations.
- Validation & Exception Center: full-dataset validation with 250 records, current exception counts, status, issues, and human-review messages.
- AI Worker Logic: Input → Analysis → Classification → Priority → Recommendation → Validation → Proceed / Human Review.

17. Submission Accuracy Note

- The intentional failure scenario used in the dashboard and demo is EKO10250. EKO99999 is not used as the current intentional failure record and should not be referenced as such.
- The current prototype should be presented honestly as a rule-based AI-assisted workflow rather than a production LLM agent. The future LLM layer is proposed, not claimed as already implemented.